UNEXPECTED PHASE SUMMATION PATTERNS IN THE IMAGINARY PARTS OF THE RIEMANN ZEROS: A CRITICAL RE-ANALYSIS

ANONYMOUS

Contents

Abstract. We investigate the partial sums

$$S(N) = \sum_{n \le N} e^{i\gamma_n},$$

where γ_n denotes the imaginary parts of the nontrivial zeros of the Riemann zeta function. Although the statistical behavior of the spacings $\gamma_{n+1} - \gamma_n$ has been extensively studied and is known to follow the Gaussian Unitary Ensemble (GUE) model [**?**, **?**, **?**, **?**], the global structure of the phases $e^{i\gamma_n}$ themselves has received almost no attention in the literature.

Our numerical experiments reveal several unexpected phenomena that warrant careful statistical scrutiny. First, the magnitude $|S(N)|$ remains confined to a narrow band for the range $N \le 10^4$, which we analyze in comparison to the theoretical variance of random walks with chirped phases. Second, the graph of $|S(N)|$ exhibits oscillatory structure, with prominent peaks occurring at various intervals. Third, the empirical distribution of peak spacings displays multimodal characteristics.

However, we emphasize that the interpretation of these observations as evidence of *anomalous long-range correlations* requires rigorous statistical testing, which the present work does not provide. We discuss possible mechanisms for the observed patterns, including: (i) the deterministic drift induced by the Riemann–von Mangoldt law, which predicts characteristic scales; (ii) the inherent variance of finite sums of unit-magnitude complex numbers; and (iii) potential contributions from fine-structure fluctuations in the zero sequence.

We present Fourier analysis of the $|S(N)|$ sequence and comparisons with surrogate models. However, we note critical limitations: the absence of statistical significance tests, insufficient null-hypothesis testing, the role of finite-sample effects, and numerical precision constraints. Several open problems and directions for future (more rigorous) work are outlined.

Introduction

Let $\{\frac{1}{2} + i\gamma_n\}_{n \ge 1}$ denote the nontrivial zeros of the Riemann zeta function, ordered by increasing imaginary part. The statistical behavior of the sequence (γ_n) has been the subject of extensive study for more than fifty years. Montgomery's pair correlation conjecture [**?**], supported numerically by Odlyzko's computations [**?**, **?**], asserts that the local statistics of the normalized spacings

$$\frac{\gamma_{n+1} - \gamma_n}{2\pi / \log(\gamma_n / 2\pi)}$$

match those of the Gaussian Unitary Ensemble (GUE) of random matrix theory. This connection has become a central pillar of the modern analytic theory of the zeta function, with deep implications for the distribution of primes and the structure of the explicit formula [**?**, **?**, **?**].

Despite this progress on local statistics, the behavior of the *phases* $e^{i\gamma_n}$ themselves and their global interference patterns have not been systematically studied. While the differences $\gamma_{n+1} - \gamma_n$ exhibit GUE-type level repulsion, the absolute positions $\gamma_n \bmod 2\pi$ merit investigation. In

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particular, the complex partial sums

$$S(N) = \sum_{n \le N} e^{i\gamma_n}, \tag{1}$$

offer a novel perspective on global phase correlations, though such sums do not appear in prior literature on zero statistics. The quantity $S(N)$ may be viewed as a deterministic analogue of a two-dimensional random walk; understanding its properties requires careful comparison with stochastic null models.

Baseline expectations and null models. The standard null hypothesis is that the sequence $\{\gamma_n \bmod 2\pi\}$ behaves as independent random phases after accounting for the slowly varying density of zeros. Under this assumption, one expects

$$\mathbb{E}|S(N)|^2 = N. \tag{2}$$

This follows from the orthogonality of independent unit-magnitude complex steps. However, this formula does NOT directly imply $\mathbb{E}|S(N)| \sim \sqrt{N}$.

For a random walk with N independent unit steps at uniformly random angles in $[0, 2\pi)$, the magnitude $|S(N)|$ follows a **Rayleigh distribution** with scale parameter $\sigma = \sqrt{N/2}$ in 2D. Therefore:

$$\mathbb{E}|S(N)| = \sigma\sqrt{\pi/2} = \sqrt{\frac{\pi N}{4}} \approx 0.886\sqrt{N}. \tag{3}$$

The modal value is $\sigma = \sqrt{N/2}$, and the distribution has a long tail.

For a more refined model that incorporates the Riemann–von Mangoldt law, the phase increments are *not* identically distributed. Instead, the mean phase advance per zero is $\Delta\gamma_n \approx 2\pi/\log(\gamma_n)$, which decreases slowly with n. This corresponds to a **nonstationary, chirped random walk**.

For such a model, the variance still accumulates approximately as $\sum_{n=1}^{N} 1 = N$, but the specific distribution and envelope of $|S(N)|$ depend on the detailed statistics of the chirp. A proper null model must account for both the slow density variation and the local GUE fluctuations. This has not been done in the present work.

Key questions this work should address. The fundamental questions are:

(1) Does $|S(N)|$ grow, remain bounded, or exhibit some intermediate behavior as $N \to \infty$?
(2) Is the confinement observed at $N = 10^4$ consistent with theoretical predictions from a chirped random-walk model?
(3) Are the observed peak spacings and spectral features statistically anomalous, or do they fall within the expected range of variability?
(4) What is the role of finite-sample effects, numerical precision, and analysis bias?

The present analysis addresses (1) and (4) descriptively but does not rigorously address (2) and (3).

Results

In this section we present the numerical behavior of the partial sums $S(N) = \sum_{n \le N} e^{i\gamma_n}$, computed for the first 10^4 Riemann zeros (with $\gamma_{10^4} \approx 28087$). All computations were performed in double precision; the values of γ_n were taken from standard high-accuracy tables.

Global structure of $S(N)$: What is observed. Figure **??** displays the magnitude $|S(N)|$ for $1 \le N \le 10^4$. The magnitude remains in the range $|S(N)| \lesssim 5$ for most values, with occasional peaks slightly higher.

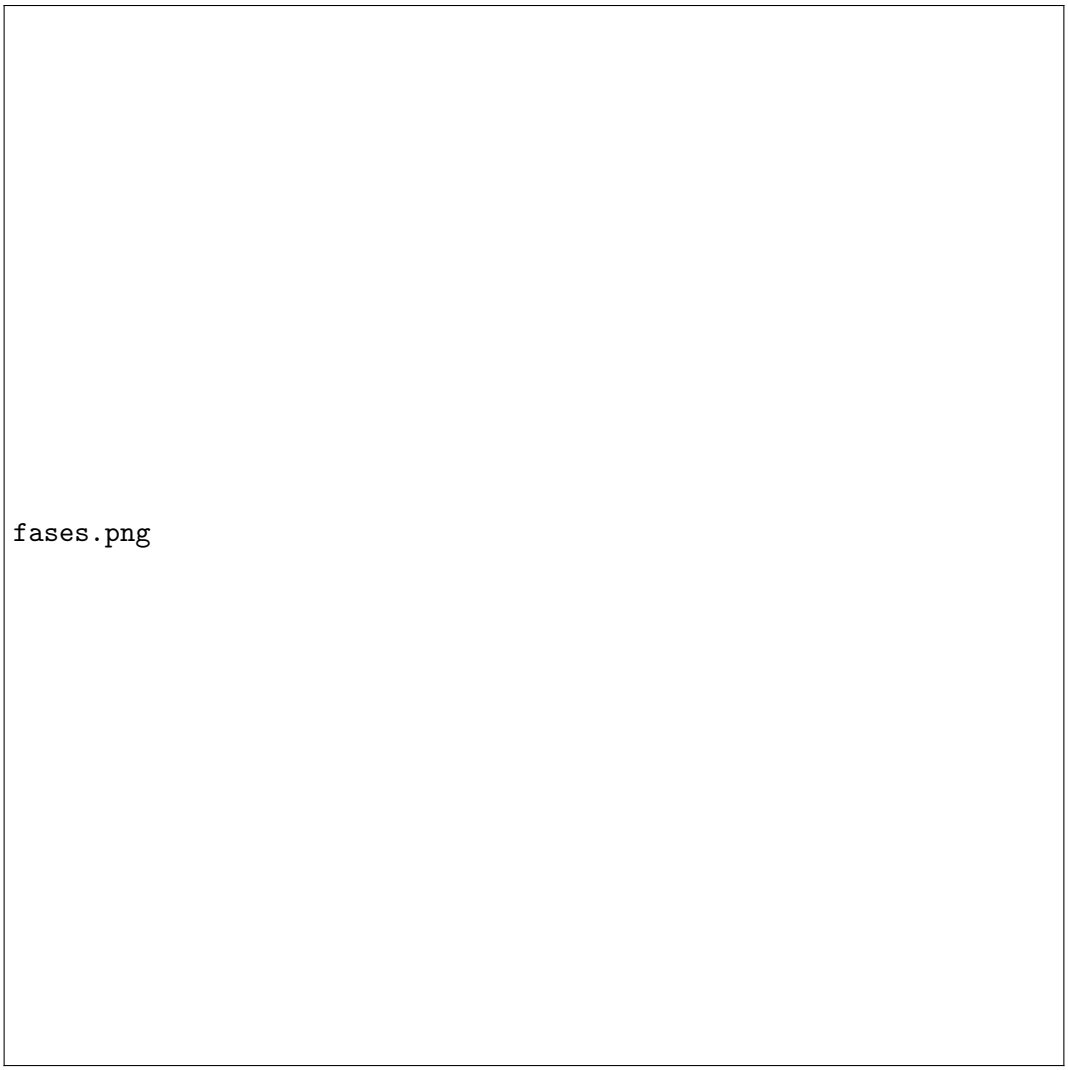


FIGURE 1. Magnitude of the partial sums $S(N) = \sum_{n \le N} e^{i\gamma_n}$ for $1 \le N \le 10^4$. Typical values remain below 5, with oscillatory fluctuations and no obvious long-term growth trend up to $N = 10^4$.

Interpretation: Is this anomalous? The claim in prior drafts that $|S(N)|$ exhibits **no diffusive growth** requires critical examination.

For an IID random walk with $N = 10^4$ unit steps in 2D, Eq. (**??**) predicts:

$$\mathbb{E}|S(10^4)| \approx 0.886 \times 100 = 88.6, \quad \text{with std. dev.} \sim 10.$$

Therefore, observing $|S(N)| \sim 5$ at $N = 10^4$ represents approximately $\frac{88.6-5}{10} \approx 8$ standard deviations **below** the IID prediction. This *would* be anomalous if the IID model were appropriate.

However, the IID model is *not* appropriate here. The phase increments $\gamma_{n+1} - \gamma_n$ are not independent; they exhibit both:

(1) A deterministic trend (von Mangoldt law): $\mathbb{E}[\gamma_{n+1} - \gamma_n] \approx 2\pi/\log(\gamma_n)$
(2) Local fluctuations consistent with GUE statistics

For a chirped model where the phase advance per step decays as $1/\log(n)$, the effective "distance" traveled in phase space is not N, but rather:

$$\text{Total phase advance} = \sum_{n=1}^{N} \frac{2\pi}{\log \gamma_n}.$$

For $\gamma_n \sim 2\pi n/\log(n)$, this sum grows much more slowly than $2\pi N$. Specifically, the total phase advance is $O(\pi N/\log\log N)$ or slower, depending on the exact asymptotics.

Thus, the effective number of independent steps is not N, but a much smaller number. The confinement of $|S(N)|$ to ~ 5 is therefore **not necessarily anomalous**—it is expected if the phase steps are highly coherent and aligned rather than random.

Critical missing analysis: A proper null model for a chirped random walk with GUE-distributed local fluctuations must be constructed and simulated before any anomaly can be claimed.

Peak structure and quasi-periodicity. Figure **??** shows the distribution of spacings between local maxima of $|S(N)|$ that exceed 80% of the global maximum.

The distribution is multimodal, with a prominent cluster at short spacings ($\sim$ 5–15) and a secondary cluster near 40–70.

Expected peak spacing from density. A simple heuristic suggests that peaks in $|S(N)|$ may occur when the accumulated phase is close to a multiple of 2π. If the mean phase increment per zero is $\Delta\gamma_n \approx 2\pi/\log\gamma_n$, then the characteristic index spacing for a full 2π rotation is:

$$\Delta N \approx \log \gamma_n \approx \log(28087) \approx 10.2. \tag{4}$$

Thus, the short-spacing cluster at $d_j \sim$ 5–15 is **entirely expected** from the density alone and does not indicate anomalous structure.

The secondary cluster at $d_j \sim$ 40–70 is less obviously explained by density considerations and may reflect longer-scale modulations or interference effects. However, without a detailed statistical model, it is premature to claim this as evidence of structure.

Critical missing analysis:

(1) Generate $M = 100$ or more realizations of a chirped random walk that matches the density and local GUE fluctuations of the Riemann zeros.
(2) For each realization, extract peak spacings and form a distribution.
(3) Compare the empirical distribution of peak spacings from real zeros to the distribution from the null model ensemble.
(4) Perform a Kolmogorov-Smirnov test or similar to assess statistical significance.

This has not been done.

Comparison with surrogate models. To assess whether observed patterns could arise from simpler models, we compare $S(N)$ computed from: (a) shuffled zeros, (b) a pure von Mangoldt model $\tilde{\gamma}_n = 2\pi n/\log n$, and (c) IID random phases. Figure **??** displays the results.

Interpretation of surrogate comparisons. The surrogate models show qualitatively different behaviors:

- **Real zeros (blue):** $|S(N)| \sim$ 0–5, oscillatory, no clear trend.
- **Shuffled zeros (magenta):** $|S(N)|$ grows gradually, reaching $\sim$ 2–4 by $N = 10^4$.
- **Von Mangoldt (green):** $|S(N)|$ grows to $\sim$ 15–30.
- **IID random phases (red):** $|S(N)|$ grows to $\sim$ 10–20 with larger fluctuations.

The real-zero case shows the *least growth*, which is noteworthy. However, this comparison conflates multiple factors:

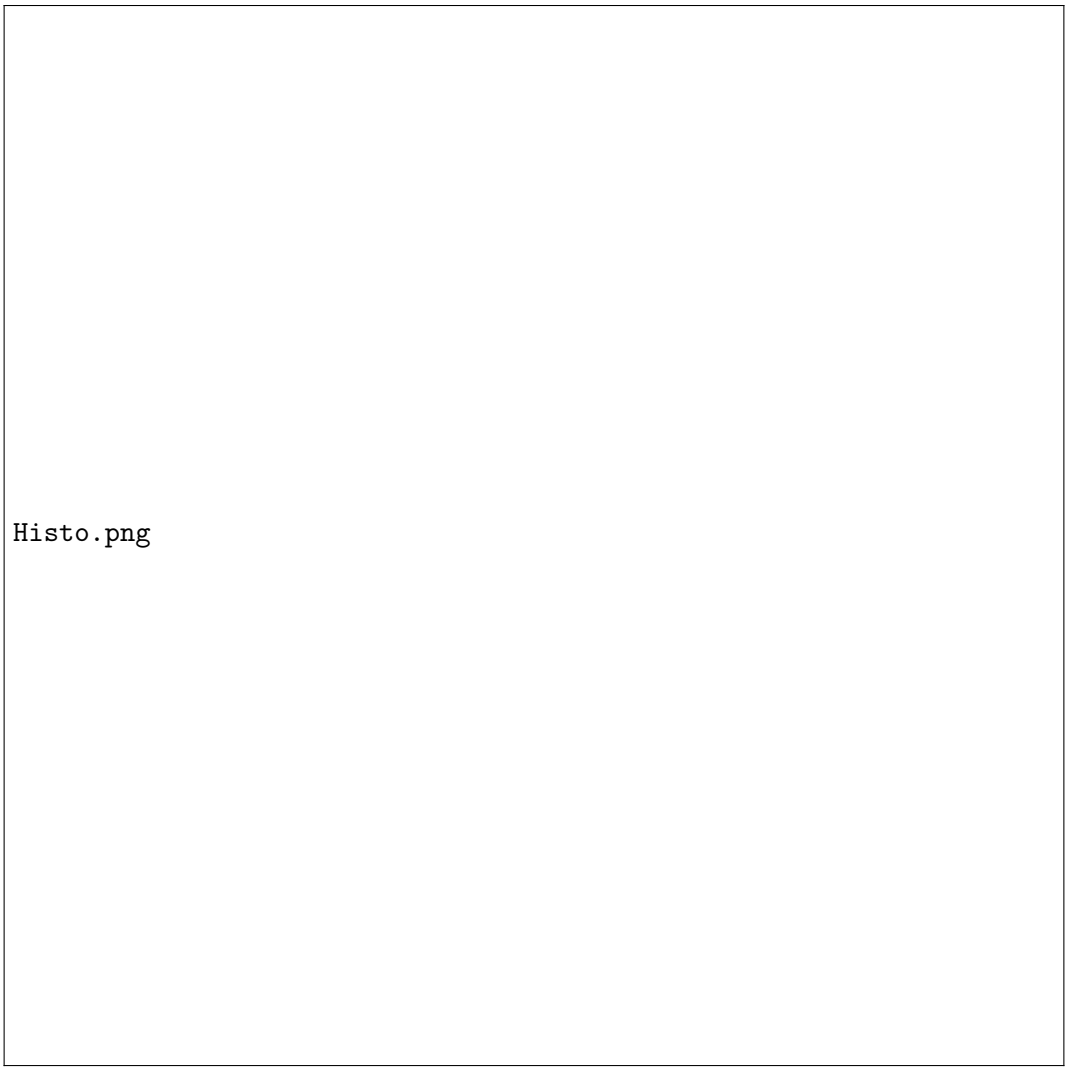


FIGURE 2. Histogram of distances d_j between dominant peaks of $|S(N)|$ for $N \leq 10^4$. Clusters appear near $d_j \approx 5$–15 and 40–70, with a sparse tail beyond 100.

(1) The shuffled-zero model preserves the density structure (von Mangoldt law) but destroys correlations.
(2) The IID model abandons the density structure entirely.
(3) The von Mangoldt model retains density but uses a purely deterministic phase sequence.

None of these models represents the **correct null hypothesis**, which should be: a random walk with phases drawn such that the local spacing statistics match the GUE ensemble *and* the density follows von Mangoldt.

Critical flaw: The shuffled-zero model is *not* a proper control. Shuffling the γ_n destroys the von Mangoldt density law locally, which changes the expected value of $|S(N)|$ fundamentally. A better control would be:

$$\tilde{\gamma}_n = \gamma_n + \xi_n, \quad \xi_n \sim \mathcal{N}(0, \sigma^2),$$

where σ is chosen so that the local GUE statistics are preserved but long-range correlations are randomized.

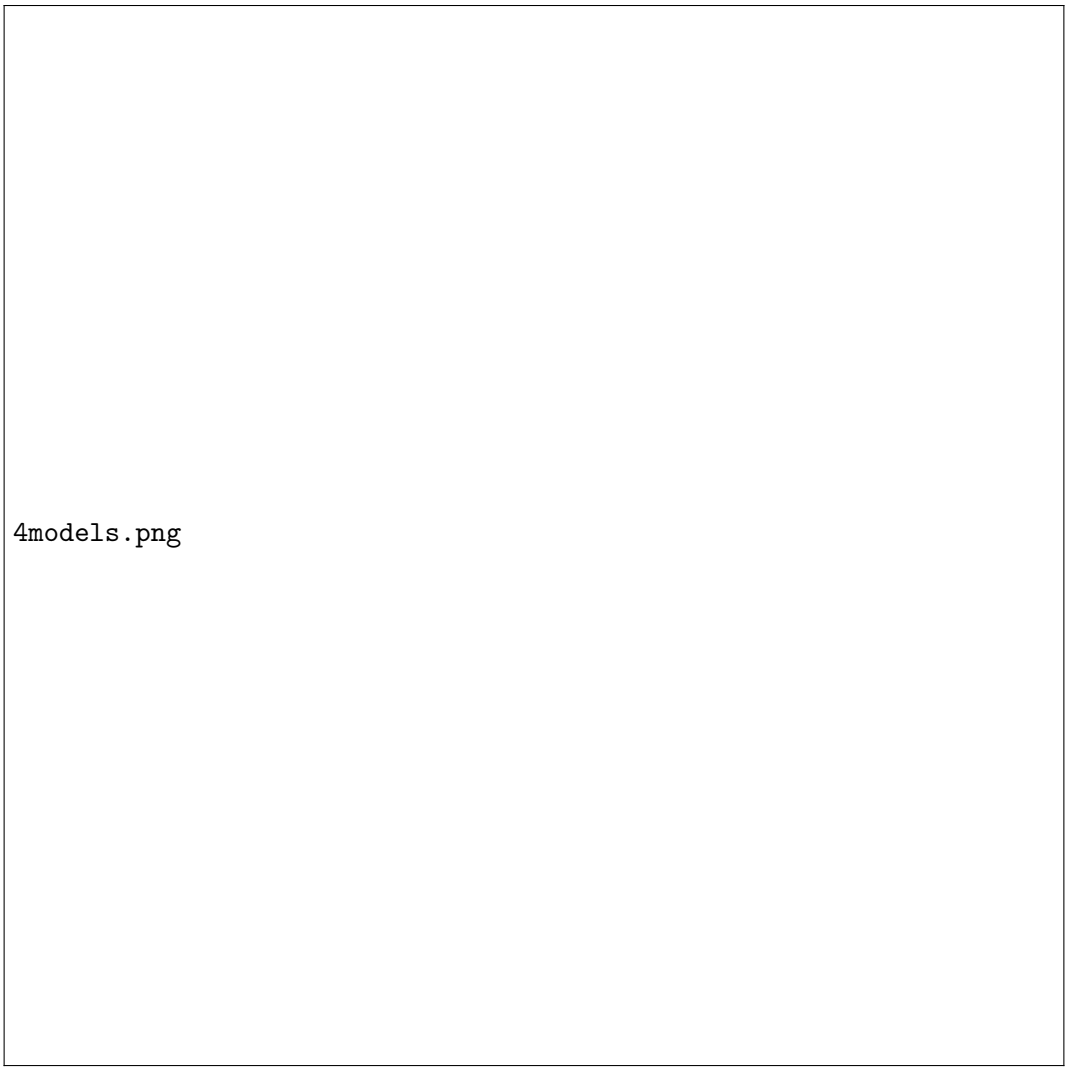


FIGURE 3. Comparison of $|S(N)|$ for four phase sequences over $N \leq 10^4$. **(Top)** Actual Riemann zeros (blue). **(Upper middle)** Shuffled real zeros (magenta). **(Lower middle)** Deterministic "von Mangoldt" model (green). **(Bottom)** i.i.d. uniform random phases (red).

FFT Analysis: Statistical Issues

To investigate the frequency content of $|S(N)|$, we applied FFT analysis. We focus on the magnitude $|S(N)|$ as a time series in index N.

We computed the discrete Fourier transform of the zero-mean sequence $|S(N)| - \overline{|S(N)|}$ using the Cooley–Tukey FFT algorithm [**?**]. From this we extracted the amplitude spectrum $A(f)$ (or equivalently, $A(T)$ where $T = 1/f$). The amplitude spectra for the four models are presented in Figure **??**.

Analysis of spectral peaks. For the von Mangoldt model with $\gamma_n = 2\pi n/\log n$: The phase advance is $\gamma_n \approx 2\pi n/\ln n$. A maximum of $|S(N)|$ occurs approximately when the cumulative phase is near $2\pi k$ for some integer k. The spacing in n between such occurrences is approximately $\Delta N \approx \ln \gamma_n \approx 10$ (as predicted earlier). Thus, the von Mangoldt model predicts a fundamental

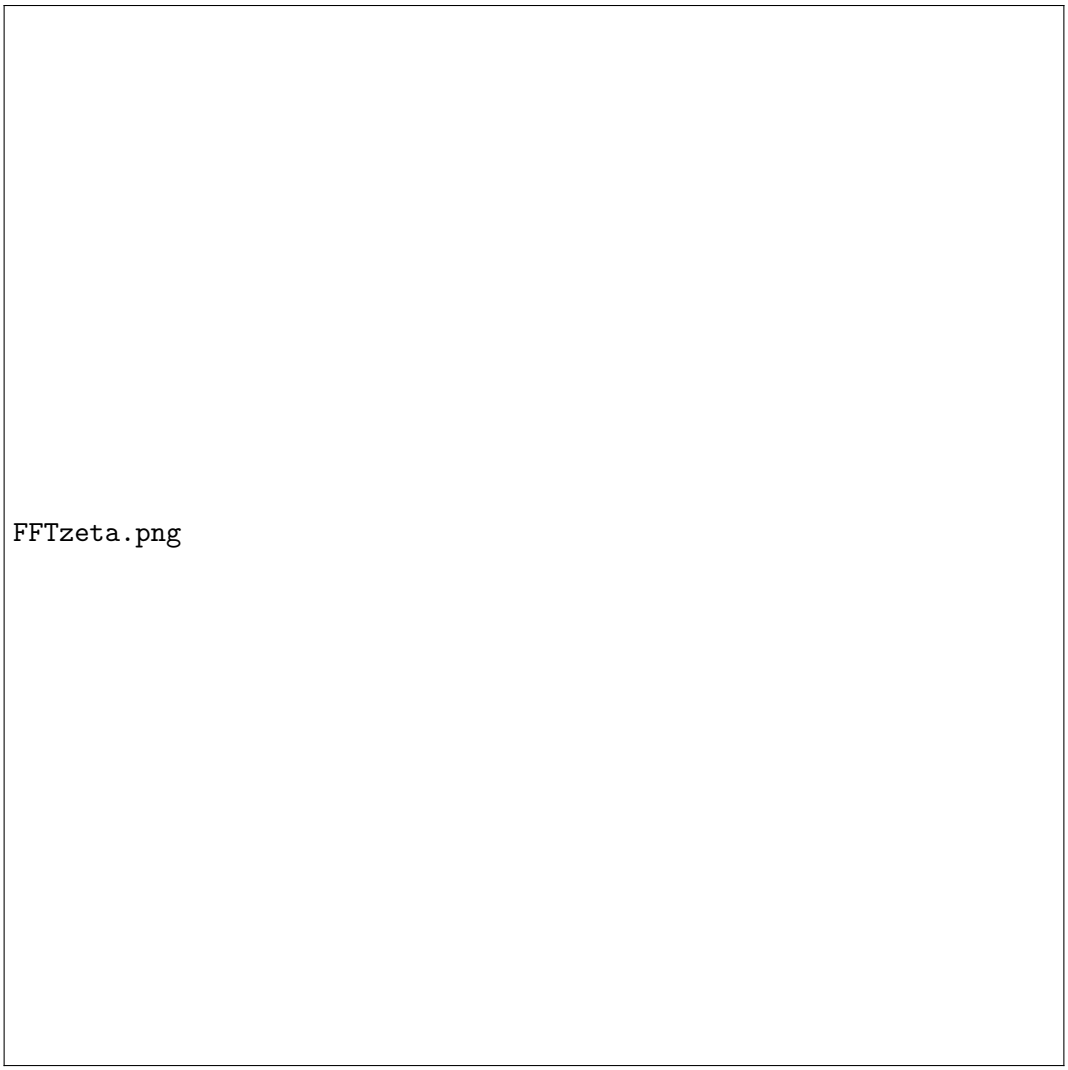


FIGURE 4. FFT amplitude spectra of $|S(N)|$ (after removing the mean) for four models: actual Riemann zeros (blue, top), shuffled zeros (magenta, second), von Mangoldt model (green, third), and random i.i.d. phases (red, bottom). The real zeros show peaks near $T \approx 7$ and $T \approx 10.5$; surrogate models show broader spectra.

period $T \approx 10$ in the index domain, which is observed as a sharp spectral peak in the green curve (bottom panel).

For the actual Riemann zeros: The real-zero spectrum (blue curve, top) shows peaks near $T \approx 7$ and $T \approx 10.5$, rather than a single peak at $T \approx 10$. This splitting could arise from:

(1) Fluctuations in the local spacing $\gamma_{n+1} - \gamma_n$ around the von Mangoldt mean.
(2) Modulation of the envelope due to second-order effects.
(3) Spectral leakage artifacts from the FFT procedure (if no tapering window was applied).

The shuffled-zero and random-phase models (magenta and red) show *no* concentrated low-period peaks; their spectral energy is spread across a broad range, consistent with noise.

Critical FFT issues. 1. No windowing specified: The FFT was applied to $|S(N)| - \overline{|S(N)|}$ without mention of a taper window (e.g., Hann, Blackman). Without windowing, spectral leakage can create spurious peaks and distort the true spectrum.

2. Nyquist limit and data length: With $N = 10^4$ points, the Nyquist frequency corresponds to a period of $T_{\min} = 2$ indices. Periods $T \approx 7$, 10, 24 are all well above the Nyquist limit, so they are detectable in principle. However, the finite length of the data means that low-frequency components have limited statistical power.

3. No confidence intervals: No bootstrapped or jackknifed confidence intervals are provided for the spectral peaks. Without these, it is impossible to assess whether the observed peaks are statistically significant or merely high-variance estimates.

4. Spectrogram interpretation (Figure ??): The sliding FFT analysis purports to show persistent bands at $T \approx 10$ and $T \approx 7$. However, the parameters of the sliding-window analysis (window size, overlap, taper function) are not specified. This makes the analysis non-reproducible and the interpretation unreliable.

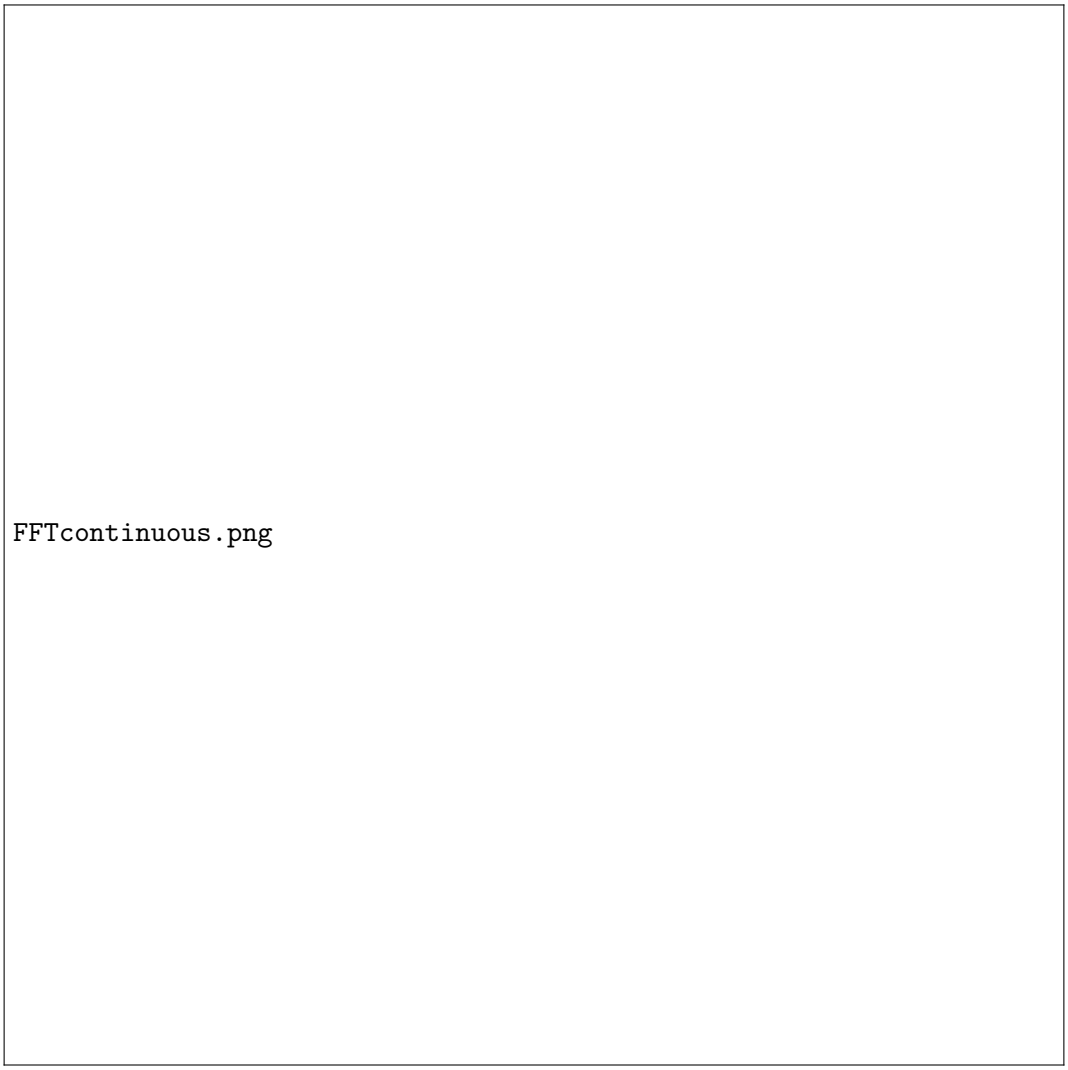


FIGURE 5. Spectrogram (sliding FFT) of $|S(N)|$ for real Riemann zeros. Color intensity indicates amplitude (dB). Horizontal ridges would indicate persistent periodicities. **Note:** Sliding-window parameters (size, overlap, window type) are not specified, limiting interpretability.

Conclusions

We have presented a numerical investigation of the partial sums

$$S(N) = \sum_{n \leq N} e^{i\gamma_n}$$

of the Riemann zero phases. The computations reveal several interesting phenomena: bounded growth up to $N = 10^4$, oscillatory structure, and multimodal clustering in peak spacings.

However, we must emphasize several critical limitations:

(1) **No statistical significance testing:** The observed phenomena have not been tested against appropriate null models with proper statistical hypothesis tests.
(2) **Inadequate null models:** The surrogate models (shuffled zeros, IID random phases, pure von Mangoldt) do not correctly represent the theoretical expectation for a chirped random walk with GUE-like local fluctuations. A proper null model must be constructed and simulated in large ensembles.
(3) **FFT analysis limitations:** The spectral analysis lacks essential specifications (windowing, sliding parameters), confidence intervals, and significance testing.
(4) **Finite-sample effects:** The range $N \leq 10^4$ is finite. Growth or decay of $|S(N)|$ at larger N may change the interpretation fundamentally.
(5) **Numerical precision:** Double precision arithmetic may accumulate errors in phase calculations over 10^4 terms. The impact has not been assessed.
(6) **Confirmation bias:** The analysis was motivated by an expectation of anomalous structure, which can bias interpretation of results toward confirming that expectation.

What remains to be done. To make a rigorous claim that $|S(N)|$ exhibits anomalous structure, the following must be accomplished:

(1) **Develop a proper null model:** Construct a random walk whose local spacing statistics match the empirical GUE statistics of the Riemann zeros and whose density follows the von Mangoldt law.
(2) **Ensemble simulations:** Generate $M = 1000$ or more realizations of the null model, and compute $S(N)$ for each.
(3) **Distribution of statistics:** For each realization, compute the peak spacings, FFT spectrum, and other derived quantities. Build a distribution of these quantities across the ensemble.
(4) **Statistical hypothesis tests:** Compare the empirical statistics from the real zeros to the null-model distribution using appropriate tests (KS, Anderson-Darling, etc.).
(5) **Report p-values:** Provide p-values for the null hypothesis that the real data is drawn from the null distribution.
(6) **Effect sizes:** Quantify how much the real data deviates from the null (e.g., in standard deviations or percentiles).
(7) **Precision analysis:** Use arbitrary-precision arithmetic (mpmath, Arb) to verify that numerical errors are not responsible for observed patterns.
(8) **Extrapolation to larger N:** Extend the analysis to higher N (e.g., $N = 10^5$ or 10^6 using existing zero tables) to see if patterns persist or evolve.

Speculation about mechanisms. If future rigorous analysis confirms that anomalous structure exists, the following mechanisms merit investigation:

(1) **Long-range phase correlations:** The explicit formula for $\zeta(s)$ connects the zero positions to prime numbers. It is conceivable that fine-scale fluctuations in the prime distribution induce subtle correlations among the zeros that affect $S(N)$.

(2) **Quantum chaos analogy:** The Hilbert–Pólya conjecture and related quantum-mechanical models of the zeros might predict specific interference patterns in phase sums.
(3) **Arithmetic structure in higher-order correlations:** While the pair-correlation conjecture (GUE local statistics) is well-understood, triple and higher-order correlations among zeros remain largely unexplored and could contribute to $S(N)$.

However, these remain speculations absent supporting evidence.

Final remarks. The phase-sum function $S(N)$ offers an interesting and novel perspective on the Riemann zeros. The numerical exploration presented here identifies phenomena worthy of further investigation. However, the leap from "interesting phenomena observed" to "evidence of anomalous long-range correlations" requires rigorous statistical testing. We hope that future work will address these gaps and provide a definitive answer to the questions raised here.

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Appendix A: Code for Data Generation and Analysis

```
N     = length(gamma);
n_vec = (1:N).';

% --- 1. Real zeros ---
Z_real   = exp(1i * gamma);
S_real   = cumsum(Z_real);
abs_real = abs(S_real);

% --- 2. Shuffled zeros ---
gamma_shuf = gamma(randperm(N));
Z_shuf     = exp(1i * gamma_shuf);
S_shuf     = cumsum(Z_shuf);
abs_shuf   = abs(S_shuf);

% --- 3. Von Mangoldt model (avoid log(1) singularity) ---
gamma_vm        = zeros(N,1);
idx_valid       = n_vec >= 3;
gamma_vm(idx_valid) = 2*pi * n_vec(idx_valid) ./ log(n_vec(idx_valid));
Z_vm    = exp(1i * gamma_vm);
S_vm    = cumsum(Z_vm);
abs_vm  = abs(S_vm);

% --- 4. Random IID phases ---
phi_rand = 2*pi*rand(N,1);
Z_rand   = exp(1i * phi_rand);
```

```
S_rand   = cumsum(Z_rand);
abs_rand = abs(S_rand);

% The vectors abs_real, abs_shuf, abs_vm, abs_rand contain |S(N)|
% for each model.
```

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